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B.Tech. Degree VII Semester Special Supplementary Examination April 2025

ME 19-205-0702 VIBRATION AND NOISE CONTROL (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Model a given physical system using spring, mass, and damping elements.

CO2: Formulate the governing differential equations of single dof, multi dof and continuous systems.

CO3: Solve the governing equations and interpret the results.

CO4: Understand the effects of noise and the control measures adopted.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A(Answer *ALL* questions)

		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a) A 0.5 kg mass attached to a light spring elongates it by 8 mm. Determine the natural frequency of the system.		3	L1	1	1,2,3
	(b) What is meant by rotating unbalance? Explain.		3	L2	1	1,2,3
	(c) What is meant by mode shapes in a multi degree of freedom system?		3	L1	2	1,2,3
	(d) Describe a method for finding the location of node in a two rotor torsionally vibrating system.		3	L2	2	1,2,3
	(e) Derive the wave equation, considering axial vibration of rods.		3	L2	3	1,2,3
	(f) What is meant by critical speed of a rotating shaft? Explain.		3	L2	3	1,2,3
	(g) Differentiate between loudness and frequency.		3	L1	4	1,2
	(h) Explain permanent and temporary hearing loss from continuous noise.		3	L1	4	1,2

PART B

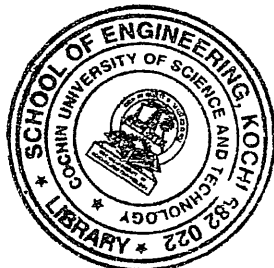
(4 × 12 = 48)

- II. (a) What is meant by logarithmic decrement? Explain. 4 L1 1 1,2,3
- (b) Determine the logarithmic decrement and the ratio of any two consecutive amplitudes of a vibrating system, which consists of a mass of 3.5 kg, a spring stiffness 2.5 N/mm and a damper of damping coefficient 0.02 N/(mm/s). 8 L3 2 1,2,3

OR

- III. (a) What is meant by Coulomb damping? Explain. 4 L1 1 1,2,3
- (b) A machine weighing 3.5 kg vibrates in a viscous medium. A harmonic exciting force of 40 N acts on the machine and produces a resonant amplitude of 18 mm with a time period of 0.2 second. Determine the damping coefficient. 8 L3 2 1,2,3

(P.T.O)



		Marks	BL	CO	PO
IV.	Setup the equations of motion of a double pendulum in terms of angle θ_1 and θ_2 , measured from the vertical. Mass $m_1 = m_2 = m$ and length $L_1 = L_2 = L$. Also determine the natural frequencies and mode shapes.	12	L3	2	1,2,3
OR					
V.	Three rotors A , B and C having moments of inertia of 2000, 6500 and 3500 kgm ² respectively are carried on a uniform shaft of 0.35 m diameter. The length of the shaft between the rotors A and B is 6 m and between B and C is 32 m. Determine the natural frequencies of torsional vibrations. Take $G = 80 \text{ GN/m}^2$ for the shaft material.	12	L4	2	1,2,3
VI.	The following data relate to a shaft held in long bearings. Length of shaft = 1.2 m Diameter of shaft = 14 mm Mass of a rotor at midpoint = 16 kg Eccentricity of centre of mass of rotor from the axis of rotor = 0.4 mm Modulus of elasticity of shaft material = 200 GN/m ² Permissible stress in shaft material = 70 MPa Determine the critical speed of the shaft and the range of speed over which it is unsafe to run the shaft. Assume the shaft to be weightless.	12	L4	3	1,2,3
OR					
VII.	Determine the frequency equation of a uniform beam having transverse vibrations for the following two cases (i) both ends simply supported (ii) both ends fixed.	12	L3	3	1,2,3
VIII.	Explain the following terms as applied to noise: (i) Sound Pressure Level (ii) Sound Intensity Level (iii) Sound Power Level (iv) Sound level meter scales (v) Psychophysical indices (vi) Acoustic spectra.	12	L2	4	1,2
OR					
IX. (a)	What are the recommended noise exposure limits by Occupational Safety and Health Administration? Plot the variation of noise exposure in decibel versus time.	6	L1	4	1,2
(b)	Six machines operating individually, make sound pressure levels of 81, 78, 79, 72, 68 and 65 dB respectively. Determine the overall sound pressure level when all of them operate simultaneously.	6	L3	4	

Bloom's Taxonomy Levels

L1 – 21.7%, L2 – 20%, L3 – 38.3%, L4 – 20%.
